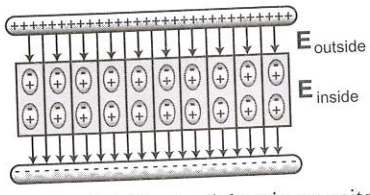


**Figure 6.7** Simple dielectric capacitor.

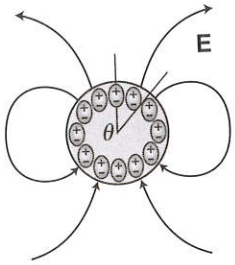


**Figure 6.8** Uniformly polarized dielectric sphere.

and the capacitance increases. Since the voltage drops while the charge on the plates of the capacitor remains the same, the energy of the capacitor decreases as the dielectric is inserted. Thus, the electric field does work on the dielectric, pulling it into the region between the plates.

As another example, we examine the electric field of a uniformly polarized sphere of radius $R$, as shown in Figure 6.8. The polarization $\mathbf{P}$ is assumed to be oriented in the $\hat{\mathbf{z}}$ direction. At the surface of the dielectric the polarization contributes a bound surface-charge density

$$\sigma_b = \hat{\mathbf{n}} \cdot \mathbf{P} = P \cos\theta \tag{6.66}$$

where $\hat{\mathbf{n}}$ is a unit vector normal to the surface of the sphere and $\theta$ the angle from the direction of polarization. This charge distribution creates a field outside and inside the sphere, as shown in the figure. To find this field we proceed as follows. We imagine that the material is composed of two equal and opposite, uniform charge distributions (the electrons and the nuclei), each one the size of the sphere. If the positive charge distribution is displaced by the amount $\delta x$ relative to the negative charges, then a uniform polarization of the medium results. Outside the sphere, the electric field is the sum of the fields of the two spherical charge distributions. But the field of a spherical charge distribution is the same as the field of a point charge at the center of the sphere, so the field outside the dielectric is just the field of two point charges slightly displaced from one another. That is, the field outside the spherical dielectric is the field of a point dipole at the origin. From the discussion of multipoles in Chapter 3, with a little effort, we get

$$\mathbf{E}_{\text{outside}}(\mathbf{r}) = \frac{3\hat{\mathbf{r}}\left(\mathbf{p}\cdot\hat{\mathbf{r}}\right) - \mathbf{p}}{4\pi\varepsilon_0 r^3} \tag{6.67}$$

at point $\mathbf{r}$, where

$$\mathbf{p} = \tfrac{4}{3}\pi R^3 \mathbf{P} \tag{6.68}$$

is the total dipole moment.

At the surface of the sphere the normal component of the field is

$$\mathbf{E}_{\text{outside}} \cdot \hat{\mathbf{n}} = \frac{\mathbf{p}\cdot\hat{\mathbf{n}}}{2\pi\varepsilon_0 R^3} = \frac{2}{3}\frac{P}{\varepsilon_0}\cos\theta \tag{6.69}$$

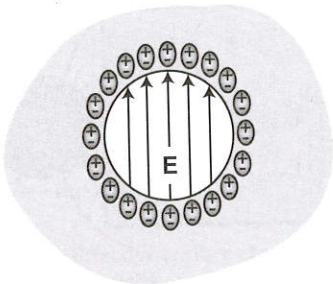


**Figure 6.9** Spherical hole in a uniformly polarized medium.

If we apply Gauss's law to the charge distribution on the surface of the sphere, we see that the field just inside the sphere satisfies the boundary condition

$$\mathbf{E}_{\text{inside}} \cdot \hat{\mathbf{n}} = -\frac{1}{3}\frac{P}{\varepsilon_0}\cos\theta \tag{6.70}$$

where the negative sign arises because the unit vector $\hat{\mathbf{n}}$ points outward from the dielectric sphere. This is satisfied by the uniform field

$$\mathbf{E}_{\text{inside}} = -\frac{\mathbf{P}}{3\varepsilon_0} \tag{6.71}$$

so the field inside the sphere is just a uniform field pointed in the direction opposite the polarization.

For a spherical cavity in a material with a uniform polarization $\mathbf{P}$, illustrated in Figure 6.9, the bound surface charge density on the surface of the sphere is opposite that of a uniformly polarized sphere with the same polarization, since $\mathbf{P} \cdot \hat{\mathbf{n}}$ has the opposite sign. Therefore, the field inside the spherical cavity is just

$$\mathbf{E}_{\text{inside}} = \frac{\mathbf{P}}{3\varepsilon_0} \tag{6.72}$$

Alternatively, we can argue that if the field throughout a uniformly polarized infinite material vanishes, then the field of the same material with a spherical (or any other shape) cavity must be everywhere exactly opposite the field that would be found from a volume of similarly polarized material having the shape of the cavity.

We can find the boundary conditions on magnetic fields in a similar manner. If we use the Gaussian surface in Figure 6.4 together with (6.24) and the divergence theorem, we find that

$$(\mathbf{B}_{\text{outside}} - \mathbf{B}_{\text{inside}}) \cdot \hat{\mathbf{n}} = 0 \tag{6.73}$$

That is, the normal component of the magnetic induction $\mathbf{B}$ is continuous. Likewise, we may use the rectangular loop of Figure 6.6 together with (6.22) to show that

$$(\mathbf{H}_{\text{outside}} - \mathbf{H}_{\text{inside}}) \times \hat{\mathbf{n}} = -\mathbf{K}_f \tag{6.74}$$

where $\mathbf{K}_f$ is the free surface-current density. In the absence of free surface currents, the tangential component of the magnetic field $\mathbf{H}$ is continuous. In terms of the magnetic induction $\mathbf{B}$ the discontinuity is

$$(\mathbf{B}_{\text{outside}} - \mathbf{B}_{\text{inside}}) \times \hat{\mathbf{n}} = -\mu_0(\mathbf{M} \times \hat{\mathbf{n}} + \mathbf{K}_f) = -\mu_0(\mathbf{K}_b + \mathbf{K}_f) \tag{6.75}$$